

Detection of Multiple Sclerosis Based on MRI Images using a Low-Power Analog Integrated Artificial Neural Networks

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Abstract—Artificial Neural Networks (ANNs) have become increasingly prevalent for classification and decision-making tasks across various engineering domains. However, their high computational resource requirements remain a significant barrier, particularly for portable or embedded applications. To address this challenge, this paper proposes an innovative method where the ANN learning phase is conducted using conventional software-based training, while the classification phase is implemented as an analog electrical circuit derived from the trained model. The method is validated on a medical diagnostic application: the detection of Multiple Sclerosis (MS) lesions from magnetic-resonance imaging (MRI) images. The dataset comprises 2,831 FLAIR and T2-weighted MRI images, partitioned into 70% training and 30% testing subsets. The proposed analog circuits achieve accuracies of 93.31% (MLP) and 94.82% (ResNet-10), while significantly reducing power consumption to 728 nW (MLP) and 967 nW (ResNet-10) and inference latency to 0.2 μ s. Although the achieved accuracy slightly trails that of purely software-based implementations, the substantial reductions in power usage and response time make this hardware-based approach highly advantageous for real-time, low-power medical diagnostic devices.

Index Terms—Multiple Sclerosis, MRI Images, Artificial Neural Networks, Low-Power Analog Systems.

I. INTRODUCTION

A. Motivation

The computational appetite of today's deep-neural-network (DNN) models has grown far faster than the energy efficiency of conventional von-Neumann hardware. A data-centre GPU such as the NVIDIA A100 sustains $\approx 3.1 \times 10^{14}$ ops s^{-1} yet draws up to 400 W, corresponding to ~ 0.8 TOPS W^{-1} — well below the power budgets of mobile and implantable devices [1]. In contrast, recent analog in-memory-computing (AIMC) accelerators have demonstrated one to two orders of magnitude better efficiency: IBM's 14 nm phase-change-memory (PCM) chip delivers 12.4 TOPS W^{-1} at software-equivalent accuracy, while the commercial Mythic M1076 analog matrix processor reaches 25 TOPS in a single 3 W package [2]. Analog hardware such as NeuRRAM and PCM-based designs achieve 10–15 $\times$ higher TOPS/W compared to

digital GPUs like the A100. By eliminating the memory-compute shuttle and exploiting physical device conductance for multiplication, AIMC hardware approaches the femtojoule-per-MAC regime envisioned by neuromorphic engineering [3].

B. Analog Neural Networks for Edge Biomedicine

Operating neural primitives in the charge or current domain allows both weighted summation and non-linear activation to occur *in situ*, reducing latency and static power. State-of-the-art AIMC prototypes such as NeuRRAM integrate 48 resistive cross-bar cores in 130 nm CMOS and measure sub-10 fJ MAC^{-1} for 256×256 vector-matrix multiplications [4]. Such energy figures translate into microwatt-scale inference engines that can be co-located with sensors, an attractive feature for continuous health monitoring.

The biomedical community has begun to harness these properties. EKGNet, a fully analog arrhythmia classifier, consumes only 11 μ W while retaining $>94\%$ balanced accuracy on the MIT-BIH dataset [5]. Likewise, a recent spiking-neuromorphic processor (Xylo) achieves real-time epilepsy detection from scalp EEG at <0.4 mW total power [6]. These examples confirm that analog neural hardware can meet the sub-milliwatt envelopes demanded by wearables and implantables without sacrificing clinical relevance.

C. Clinical Gap: Multiple Sclerosis (MS) Diagnosis

MS is a chronic autoimmune disease of the central nervous system whose progression is best tracked by magnetic-resonance imaging (MRI). Deep-learning approaches based on 2-D/3-D convolutional neural networks (CNNs) routinely exceed 90% accuracy in classifying MS lesions, yet their deployment still relies on power-hungry GPUs or cloud resources [7]. Portable MRI scanners and point-of-care analyzers therefore lack an embedded, low-latency solution for automated MS assessment.

Despite rapid advances in AIMC, no prior work has mapped an MS lesion classifier to a fully analog CMOS implemen-

tation. Existing analog chips target ECG, EEG, or keyword spotting, and software-only MS models ignore hardware constraints. This paper closes that gap by co-designing the network topology, training quantization, and circuit primitives to realise an *ultra-low-power, current-mode MS classifier* in standard CMOS. Our system integrates acquisition, preprocessing, inference, and output in a tightly coupled flow, enabling deployment in low-power embedded contexts.

D. Contributions

This work presents significant advancements in low-power analog neural network implementations for medical image analysis, specifically targeting MS lesion detection. The main contributions are summarized as follows:

- **First Analog MS Classifier:** The first fully-analog CMOS-based neural network tailored specifically for detecting MS lesions using MRI images, fabricated in a 90 nm CMOS process.
- **Efficient Hardware-Software Co-design Methodology:** Developed a quantization and mapping approach to effectively translate software-trained neural networks (MLP and ResNet-10 variants) into subthreshold analog hardware implementations.
- **Ultra-Low Power and Latency:** Achieved sub-microwatt power consumption (728 nW for MLP and 967 nW for ResNet-10) with inference latency as low as 0.2 μ s, suitable for always-on portable medical devices.
- **Robust Diagnostic Performance:** Demonstrated high accuracy (93.31% for MLP and 94.82% for ResNet-10) on extensive MRI datasets, validated under realistic process, voltage, and temperature variations.

II. MATERIALS AND METHODS

A. Dataset Selection

The dataset used in this study is the Brain MRI Dataset of Multiple Sclerosis with Consensus Manual Lesion Segmentation and Patient Meta Information [8]. It contains 2,831 FLAIR and T2-weighted MRI slices from 60 patients. Each slice is annotated with an expert-validated binary lesion mask. Patient-level metadata (EDSS scores) is included. The dataset is balanced between MS and non-MS classes. Representative examples from the dataset are shown in Fig. 1, illustrating both raw FLAIR and T2-weighted MRI slices alongside their corresponding binary lesion masks.

1) *Pre-processing Pipelines:* Two distinct pipelines were defined:

-*CNN pipeline:*

- **Scaling:** pixel values scaled to [0, 1];
- **Resizing:** images resized to 300×300 with anti-aliasing;
- **Padding:** images padded to 320×320 pixels, maintaining aspect ratio.

-*MLP pipeline:*

- **Rigid registration** (FLAIR and T2);
- **Skull-stripping;**
- **Bias-field correction;**

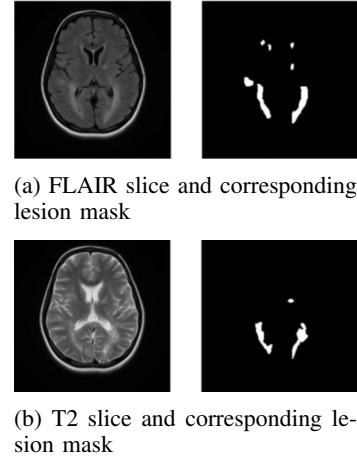


Fig. 1: Examples from the MS dataset: each row shows a raw MRI slice (left) and its binary lesion mask (right).

- **Per-subject z-score normalization,** mapped later to 4–16 nA;
- **Patch extraction:** 3D cubes ($11 \times 11 \times 11$) from lesion and non-lesion regions.

B. Neural Network Architectures

Two complementary artificial neural network (ANN) pipelines were implemented: (i) a shallow multilayer perceptron (MLP) with handcrafted 3D patch features, and (ii) a deep CNN based on ResNet-10 directly utilizing 2D MRI slices. ResNet-10 was selected due to its favorable trade-off between computational cost and classification performance, making it particularly suitable for deployment in embedded clinical systems [9]. Both networks were initially trained in software and subsequently mapped onto ultra-low-power analog hardware.

Common training settings.: All networks were trained using a 70/30 train-test split, binary cross-entropy loss, He-normal initialization, and early stopping based on validation loss.

1) MLP Implementation:

Input representation and feature engineering.: The features extracted from the volumetric patches include voxel-level statistics (mean, standard deviation), first-order texture metrics (entropy, contrast, homogeneity), and concatenated flattened intensity values. The resulting feature vectors were normalized per subject and mapped into the analog operating current range (4–16 nA).

Network architecture.: The MLP architecture includes:

- **Input layer:** determined by feature vector length ;
- **Hidden layer 1:** 100 neurons, sigmoid activation;
- **Hidden layer 2:** 100 neurons, Gaussian activation;
- **Output layer:** 50 parallel neurons outputting currents for hardware compatibility.

Training: The MLP was optimized using stochastic gradient descent (SGD), with the learning rate and epoch

count selected by grid search. The final trained weights were quantized for analog mapping.

The architecture of the MLP-based classification pipeline is illustrated in Fig. 2, where extracted 3D patches are converted to feature vectors and processed through two fully connected (FC) layers prior to classification.

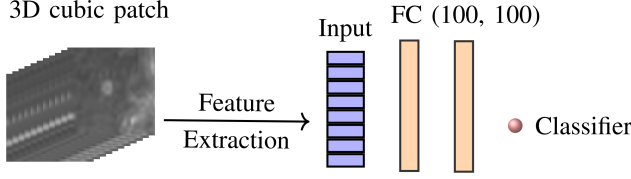


Fig. 2: Schematic of the MLP-based classification pipeline.

2) CNN Implementation:

Input representation: CNN inputs were 2D pre-processed grayscale axial MRI slices.

Baseline architecture - ResNet-10: The ResNet-10 model [10] comprises:

- **Initial Convolution:** 3×3 convolution (16 filters), Batch-Norm, ReLU;
- **Four Residual Blocks:** each with two convolutions and skip connections (16→128 filters);
- **Global Average Pooling (GAP);**
- **FC Layer:** 512 neurons, ReLU;
- **Output Layer:** single sigmoid neuron.

Compressed architecture for hardware: The network was compressed to two convolutional layers (16→32 filters), replacing GAP with an approximate current-averaging mechanism and removing the FC layer in hardware implementation.

Training: CNN was trained with Adam optimizer and data augmentation (flips, rotations $\pm 5^\circ$). The final weights were quantized for analog mapping. The complete ResNet-10 baseline architecture is illustrated in the upper part of Fig. 3, while the compressed variant is shown in the lower part.

C. Analog Hardware Mapping

1) Circuit Architecture and Power Design: To maximize power efficiency, all circuits operate with $V_{DD} = -V_{SS} = 0.3V$, and transistors are biased in the subthreshold region to minimize current draw. Each class is modeled with a block containing neuron cells that independently process two input features. The system supports N_d inputs and two output classes, as shown in Fig. 4. A group of neuron cells represents each class. The primary building blocks of the architecture process two input features independently in each neuron cell.

Initially, for each feature, the input current $I_{in,i}$, where $i \in [1, N_d]$, is applied to the sigmoid function circuit (SFC) [11] shown in Fig. 5. This circuit is composed of a NMOS Winner-takes-all (WTA) circuit which implements two sigmoid function currents based on the input and parameter currents I_{in} and I_r respectively [12]. The output currents is a sigmoid function current which is tuned by the parameter current I_r (mean value) and I_{bias} which alters the height.

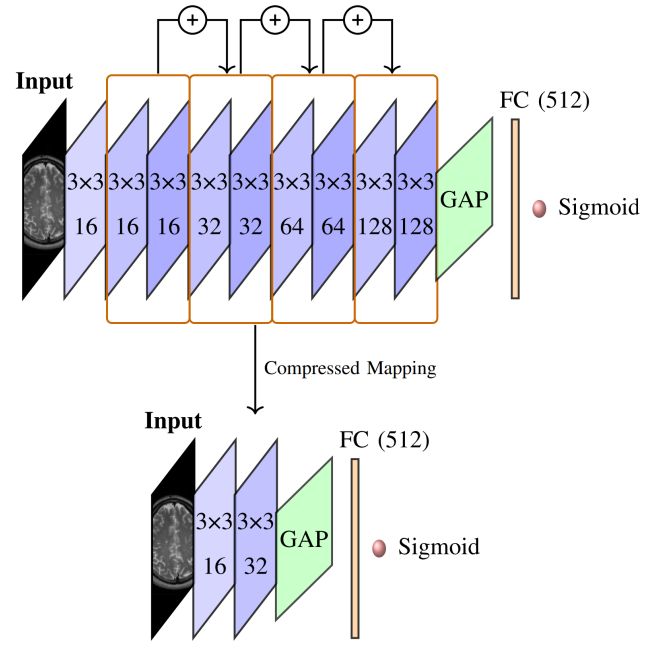


Fig. 3: Comparison between ResNet-10 and its compressed counterpart for ultra-low-power hardware implementation.

The output of the SFC, alongside the training parameters I_r , V_c and I_{bias} , are then connected to the input of the next analog building block, the Gaussian Function Circuit (SFC) [13] shown in Fig. 6. This design uses a modified version of the WTA to realize a differential pair composed of composite transistors. This architecture splits the bias current I_{bias} into sigmoid-shaped currents centered around the reference voltage (I_r). Furthermore, the control voltage V_c connected to the bodies of transistors M_{n2} and M_{n5} adjusts the curve's width, enabling tunability. The selected gaussian curve is the one transitioning from high to low and its magnitude is determined by the bias current.

An analog multiplier [14], shown in Fig. 7, receives the currents from the two SFCs of the neuron cell and multiplies them to generate the neuron's output. The translinear principle is used in this implementation to calculate the product of input currents I_a and I_b . The reference current I_{div} normalizes the generated product. The analog multiplier computes the product:

$$I_{out} = \frac{I_a \cdot I_b}{I_{div}}$$

This translinear circuit multiplies subthreshold currents I_a and I_b , normalized by a reference current I_{div} , yields the output current. The gate to source voltages inside the loop can be converted into the current product thanks to this design's subthreshold operation. The subthreshold region's current and gate to source voltage exhibit exponential behavior, which makes this design extremely effective while guaranteeing precise current multiplication [15].

The final output current of the corresponding class is then calculated by adding the currents that each neuron cell

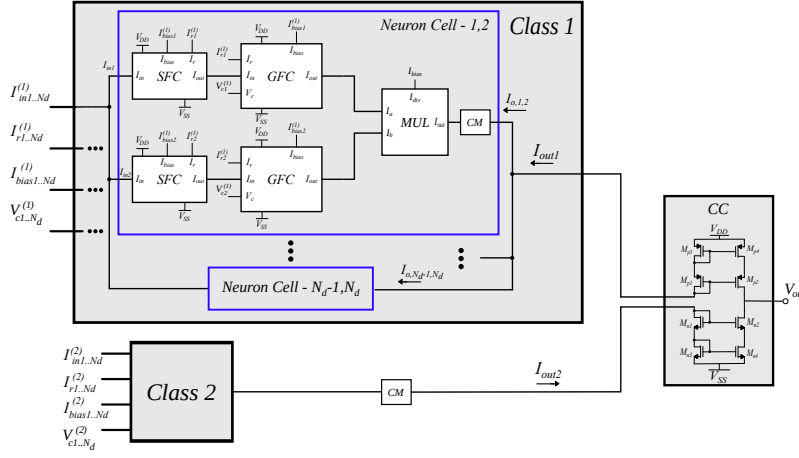


Fig. 4: The high-level architecture of the proposed analog hardware ANN.

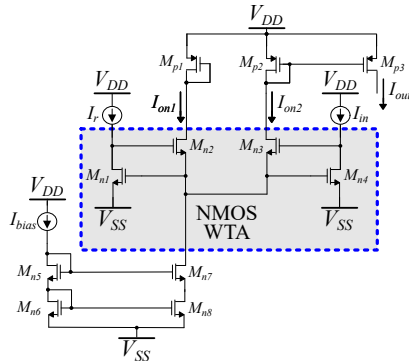


Fig. 5: The Implemented SFC.

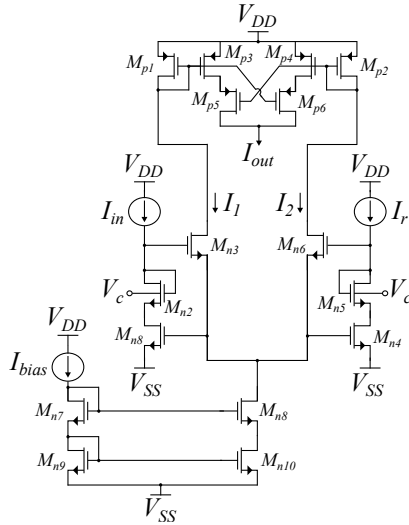


Fig. 6: The Implemented GFC.

produces. The design heavily utilizes Current Mirrors (CMs) to maintain appropriate signal routing between blocks, guaranteeing that the output current from each block flows appropriately into the subsequent stage. The final decision is made by the Current Comparator (CC) block in Fig. 4, which analyzes the

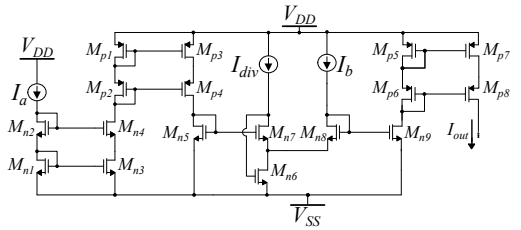


Fig. 7: The analog translinear implementation of the multiplier.

class output currents and makes V_{out} swing to high or low depending on which class dominates.

The trained network's weights and biases were mapped onto the analog circuit parameters, which are summarized in Table I and II.

TABLE I: GFC Parameter Ranges

Parameter	Range	Description
I_r	4 to 14 nA	Center of Gaussian
V_c	-0.3 to 0.3 V	Variance of Gaussian
I_{bias}	4 to 16 nA	Bias current

TABLE II: SFC Parameter Ranges

Parameter	Range	Description
I_r	4 to 18 nA	Sigmoid mean value
I_{bias}	5 to 12 nA	Sigmoid height

Due to the absence of auto-scaling hardware, a fixed I_{mul} was assigned, highlighting the trade-off between power efficiency and classification accuracy. Quantization and clipping ensured analog parameter compatibility.

Performance metrics included accuracy, power consumption and classification speed are provided. Mapping the ANN to hardware demonstrated a correlation between parameter tuning and classification reliability. Wider analog parameter ranges improved robustness but increased power consumption, underscoring a power-performance trade-off. Final network

configurations were stored in analog memory for deployment in low-power, real-time lesion classification applications.

2) Software-to-Hardware Mapping:

Shared analog parameters: Trained weights/biases are quantised into I_T , I_{bias} , V_c ranges listed in Tables I–II; a fixed multiplier bias $I_{mul}=8$ nA avoids on-chip auto-scaling.

MLP instantiation: Each hidden node maps to one SFC→GFC→Mult cell. The software sigmoid output is replaced by a 50-neuron current bank, summed and compared to I_{ref} , yielding mismatch-robust inference ($< 1\mu W$, see Sec. II-C1).

Compressed-CNN instantiation: A 3×3 current-mode cross-bar performs one-cycle convolution; folded-cascode mirrors plus a resistive ladder approximate GAP. Sixteen soft-class neurons replace the FC layer and feed the same comparator. Kernel weights fall in the 4–16 nA range and are programmed through the shared 12-bit DAC interface.

The complete classifier array was taped-out in TSMC 90 nm CMOS and occupies an active area of $1.2 \times 1.8 \text{ mm}^2$ (see Fig. 8).

We note the following measured metrics:

- **MLP total power:** 728 nW
- **Compressed-ResNet total power:** 967 nW
- **Inference latency:** 0.2 μs

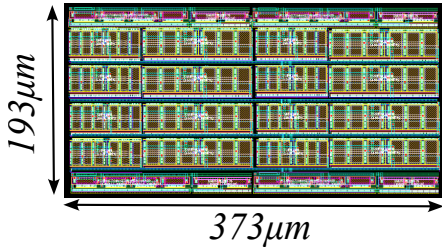


Fig. 8: The layout implementation of the proposed architecture consisting extra dummy transistors.

III. SIMULATION RESULTS

The performance of the proposed analog neural network classifier was evaluated using two primary tests for both training procedures. All experimental results were derived from the layout, shown in Fig. 8, realized through a TSMC 90nm CMOS process using Cadence Virtuoso. The first test evaluated the impact of training randomness by conducting 1000 independent training iterations for each of the training procedures used in the analog hardware implementation. Fig. 9 presents a comparative analysis of classification accuracy for both approaches.

To reduce randomness in the train-test split and obtain a more accurate estimate of the ANN classifier’s accuracy, 1000 iterations were performed (for both training procedures). We run 1000 iterations with different training-test splits using cross-validation to ensure the results are statistically reliable and to account for variability in the data partitioning. This approach helps to evaluate the model’s performance more robustly by averaging over many possible data splits. This allows

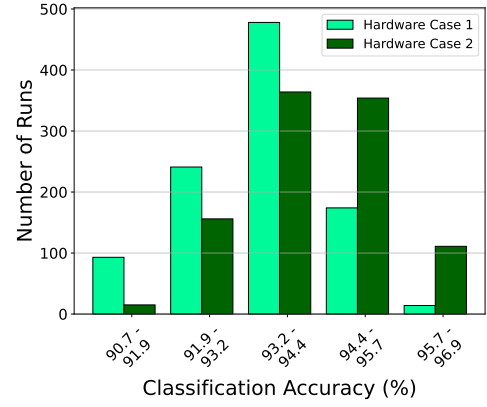


Fig. 9: Classification outcomes of both cases on the dataset across 1000 iterations. Hardware Case 1: 2-layer MLP-based architecture. Hardware Case 2: Compressed ResNet-10-based architecture.

for the calculation of a mean value and a range that reflect the model’s behavior and robustness across various scenarios, resulting in a more reliable and consistent evaluation. For each scenario, a 70–30% train-test split was used.

To evaluate robustness under varying physical conditions, a Process-Voltage-Temperature (PVT) corner analysis was conducted. The following conditions were tested:

- **Process corners:** Typical, Slow, and Fast
- **Temperature range:** from $-25^{\circ}C$ to $125^{\circ}C$
- **Voltage range:** $V_{DD} = -V_{SS} = 0.25 \text{ V}$ to 0.35 V

The worst-case performance was observed under the combination of Slow process, Cold temperature ($-25^{\circ}C$), and Low voltage ($V_{DD} = -V_{SS} = 0.25 \text{ V}$), where the classifier achieved an accuracy of 88.73% for simple ANN and 89.98% for ResNet-10 case. Even under these extreme conditions, the system maintained functional integrity.

In addition, a 5000-iteration Monte Carlo simulation was performed to account for process variations and device mismatch. Fig. 10 shows the distribution of classification accuracy under these statistical variations. The results indicate that the proposed architecture is highly robust and maintains reliable performance under a wide range of manufacturing and environmental conditions.

More specifically, we conducted 5000 Monte Carlo simulations of PVT variations (both process and mismatch) in the ANN circuit to statistically capture the impact of manufacturing and environmental variability on circuit performance. This extensive sampling ensures a robust and reliable assessment of ANN behavior under real-world conditions. The mean value is equal to 93.31% and the standard deviation is equal to 1.73% for the training procedure related to the simple ANN. The mean value is equal to 94.82% and the standard deviation is equal to 1.87% for the training procedure related to ResNet-10. These findings validate the effectiveness of the analog classifier and its suitability for deployment in energy-constrained, variable environments such as point-of-

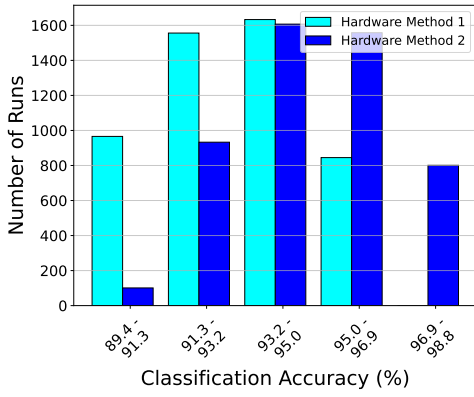


Fig. 10: Post-layout Monte-Carlo simulation results of both cases on the dataset. Hardware Method 1: 2-layer MLP-based architecture. Hardware Method 2: Compressed ResNet-10-based architecture.

care diagnostic systems.

IV. DISCUSSION AND ANALYSIS

In the previous Section, a detailed performance evaluation of the proposed analog neural network classifier, including statistical analysis across training procedures and robustness under varying conditions. Both architectures were implemented (trained) in software and tested on-chip. The two training procedures refer to: (a) training the simple ANN architecture and (b) training the ResNet-10-inspired architecture. For both cases, to ensure statistical reliability and account for training variability, 1000 independent training test iterations were performed using cross-validation for both the simple ANN and ResNet-10 training procedures. In each case, a 70–30% train-test split was applied to provide the mean and range of classification accuracy. As shown in Fig. 9, the ResNet-10 architecture consistently outperformed the simple ANN training methodology, with an average classification accuracy improvement (during the validation test) of approximately 1.5% across all training-test splits. This confirms the robust behavior during different train-test splits.

Each software-trained model (for example, simple ANN vs. ResNet-10) has a different complexity due to the number of layers, weight distribution and circuit sensitivity. Even when implemented on the same fabrication process and design methodology, the main realizations will respond differently to process variations, device mismatches and chip/environmental fluctuations. Monte Carlo analysis allows one to quantify and compare the statistical robustness of each model’s hardware implementation under identical variability conditions. In general, hardware architectures are affected by circuit-level non-idealities, including changes in the values of training parameters and chip-level fluctuations, which impact the overall circuit behavior. While the ResNet-10 case shows a higher standard deviation under Monte Carlo analysis, it also achieves better classification performance compared to the simple ANN.

V. CONCLUSION

This work has presented a novel approach for implementing analog neural network models by translating software-trained networks into optimized electrical circuits. The developed analog circuits exhibit ultra-low power consumption and provide reliable computational performance, making them highly suitable for portable medical applications. The proposed methodology was specifically demonstrated on a MS classification task using MRI images, achieving robust and accurate diagnostic performance.

Future work includes assessing the robustness of the proposed analog neural network system across additional MS-related MRI datasets. Furthermore, further optimization strategies will be explored to further reduce inference latency and enhance the overall responsiveness of the system.

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